

Activity 9

LEVELS:

Grades 9 to 12

COURSES AND EXPECTATIONS:

Canada and World Issues

SSV.02, GCV.01, GIV.01, GIV.02,
GIV.03, GIV.04, SS1.04, SS1.05,
SS3.02, GC3.4, GI1.02, GI2.02,
GI2.06, GI2.07, GI2.08, GI2.10

TASK:

Through GIS processes, students investigate the current and changing nature of the global economy.

EXPECTATIONS:

The student will:

- Explore the world economy in relation to GDP, poverty, debt, investment and income equity.
- Prepare maps showing GDP, poverty, debt, investment and income equity.
- Analyze world poverty and external debt.
- Use and query data tables linked and associated with maps.
- Compute and use new data tables.
- Display computed data using charts.
- Prepare a map layout for printing.

TIMEFRAME:

90 Minutes

ACHIEVEMENT CATEGORIES:

Knowledge / Understanding

Thinking / Inquiry

Communication

Application / Making Connections

MATERIALS:

Computer system with ArcView GIS software

VOCABULARY:

GDP

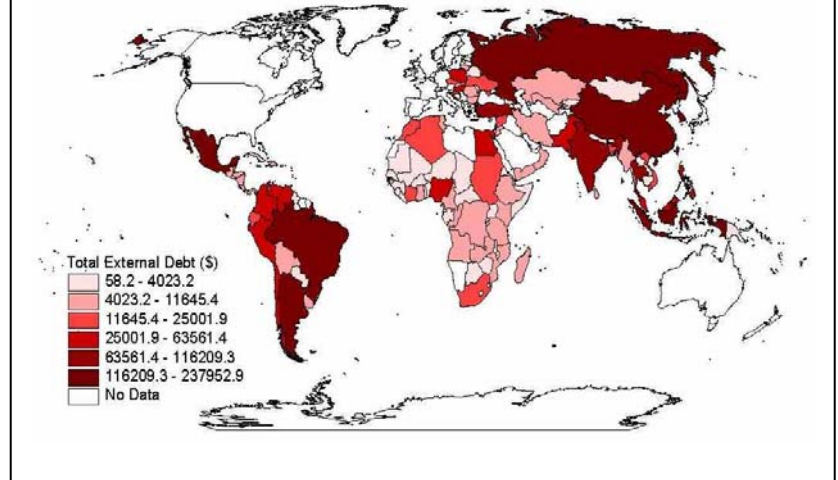
Foreign investment

Poverty

GINI index

Development assistance

Total External Debt



The Global Economy

PROCEDURE:

1. Students will research background information on global economy.
2. Students will download data sets needed for GIS analysis of the global economy. (GDP, foreign investment, poverty, income equity, development assistance) The data can be found on the website <http://www.ocdsb.edu.on.ca/woodsweb/Geomatics> and downloaded before doing the lesson.
3. Students will follow the defined GIS exercise to develop products that provide insight into the current and changing global economy.
4. Students will use the GIS output products to answer questions that are imbedded into the exercise.

DISCUSSION

Knowledge / Understanding

1. What is the global economy? Compare economic variables in several different countries (developed, developing, undeveloped).
2. How can countries move towards a sustainable economy?
3. Which countries show positive change and which countries show negative change? Why?

Application / Making Connections

4. Where do we see a sustainable economy in the real world?

Thinking / Inquiry

Start a discussion of the following:

5. What do we need to know in order to measure the global economy?
6. What is needed to improve the global economy (e.g. leadership, communication, trust, legislation, technology, trade, understanding of consequences, examples of failure, etc)
7. How does GIS analysis help to understand the complexity of the global economy?

HOMEWORK SUGGESTION:

Research paper:

1. Select an additional aspect of the economy. (E.g. Corporations, exports / imports, spending profiles, etc.)
2. Research the selected topic for 10 different countries to support a background paper and a GIS project.
3. Find, select and acquire data for the 10 countries to be used in the GIS project. Add the data to the gdp_pc.dbf for GIS products.
(http://earthtrends.wri.org/searchable_db/index.cfm?theme=10)
4. Produce GIS products (maps, charts or tables) to support the background paper.
5. Define the implications for the global economy in the 10 countries.